

Improving Relative Pose Estimation by Estimating Gravity

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Abstract

Estimating the relative pose of cameras is an important problem in 3D vision applications. It is used to create 3D models from images and for mapping in robotic applications. Especially for real-time applications, it is important to have fast and robust algorithms for relative pose estimation. For calibrated camera settings, without further constraints, the relative pose can be estimated using a 5-point solver. By using the gravity vector of the images, the problem reduces to 3 DoF. This allows to reduce the sample size and thereby decrease the required amount of RANSAC iterations exponentially. In this work we develop a pipeline to estimate the relative pose using ‘3-point (with gravity)’ and ‘5-point’ estimators, and compare their performance on the ScanNet dataset. We perform tests on more than 11000 image pairs from 260 indoor scenes and varying amount of errors in the gravity estimates.

1. Introduction and Problem Formulation

Estimation of relative camera positions from images of a scene taken from multiple view points is one of the core problems in the field of computer vision. Downstream tasks like visual localization, simultaneous localization and mapping (SLAM), and structure-from-motion depend on the efficient and reliable estimations of the camera positions. Fast and robust relative pose estimation is critical to applications like augmented-reality [26], robotics [21, 9] and autonomous navigation [3].

RANSAC is a widely used method for model fitting and it is a critical part of various computer vision pipelines. Consequently, the overall efficiency and robustness of the pipeline is dependent on the accuracy and convergence of this model fitting step. An important factor that determines the iterations required by RANSAC to converge to an accurate solution is the number of point correspondences required to test whether they agree with the current hypothesis or not. Intuitively, as the number of required correspon-

dences increases, the chance of encountering an outlier in the selected set of correspondences also increases. Hence, reducing the number of point-correspondences is of great advantage for ensuring robustness and efficiency.

The classical relative pose estimation problem from a given pair of images and known camera intrinsic parameters requires 5-point correspondences. However, with the additional constraint of a known axis of rotation, the degrees of freedom for this problem reduces by 2, and after this to get the relative pose only the rotation about the known axis and translation (up to scale) is required, which can be estimated by using just 3-point correspondences.

Gravity vector is one of the natural choices of this known axis of rotation. For applications in robotics, the gravity information is readily available with the use of IMUs. Such sensors are almost always available in personal image capturing devices like smartphones and tablets. For cases when this information is not available, the gravity vector can be estimated using a learned model such as Upright-Net [28]. Again, for most of the real-world scenes and use cases, the objects in the image provide strong cues to reliably determine the gravity vector. For example, for outdoor scenes; road surfaces, vehicles, pedestrians, trees *etc.* provide strong signals for such models to determine the gravity vector because certain planes and lines on these objects have a fixed orientation with respect to the gravity vector in natural settings. Similarly, for indoor scenes, the examples of such entities are walls and their junctions, ceilings, tables, chairs *etc.* However, using such models has additional computational cost, but for most use cases it has to be applied once per image unlike the model fitting step which typically runs on multiple image pairs.

In this work, we evaluate the 5-point based classical and 3-point-upright (using gravity) relative pose estimation algorithms on scenes from the ScanNet dataset. We compare these two methods in terms of pose errors and runtimes on indoor scenes with varying complexity. Another point to note is that the gravity information is not guaranteed to be noise-free, especially with the model estimates. Therefore, we also test the performance of 3-point-upright estimation

on estimates from the UprightNet and further with synthetically added noise to this estimate.

2. Related Work

The gravity and relative pose estimation components of our pipeline builds upon the previous works in the field of computer vision and deep learning. For gravity estimation we use the state-of-the-art gravity estimation network UprightNet [28]. Unlike previous methods which directly regress over gravity, UprightNet tries to enforce geometric constraints as well while keeping the whole process end-to-end differentiable. It predicts scene geometry representations in both the local camera and global reference coordinate systems, and then estimates the best rotation that aligns these two predictions via differentiable least squares.

The classical methods infer the gravity by relying on geometrical features [11, 18]. The orthogonal line segment in the image can be used to estimate the horizon line and thus infer the gravity vector [11, 18]. The other contributions propose gravity estimation using coplanar circles [2] and repeated patterns [24]. However, these methods depend on handcrafted design which limits its robustness towards broader applications. On the other hand, machine learning-based methods have been successfully applied to this problem and have surpassed previous methods. Other viable options to the chosen UprightNet [28] are Deephorizon [25] which specializes in outdoor panorama scenes as well as approaches based on intrinsic and extrinsic camera calibration using CNN [29, 17]. Another such model is GroundNet [16] which estimates the ground plane as the intermediate step to calculate the gravity vector. However, this method fails if there is no ground plane visible.

A further component is to use RANSAC [7] in conjunction with a solver (in our case, a 3-point or 5-point solver). RANSAC is a model fitting algorithm that can greatly improve robustness of an estimation process. It samples the minimum number of data points required to generate candidate hypotheses, and find the number of data points that agree with it (inliers). The goal of RANSAC is finding the best hypothesis that agrees with most of the input data. Even though RANSAC provides robustness, the iterative nature of RANSAC results in calculation overhead. The number of iterations can be minimized by reducing the number of data points or by reducing the noise in the data [7].

RANSAC has been widely used for relative pose estimation [1, 19, 10, 5, 6]. All of these methods use their respective relative pose solver in combination with RANSAC. As mentioned previously, the less points are required, the better will be the computational complexity of finding a plausible solution. Hence, the pose estimation pipelines which use classical 8-point algorithm [1] and 5-point algorithm [19] can be improved by reducing the number of iterations

of RANSAC by using known geometrical information (like the gravity vector) to reduce the number of minimal points required by the RANSAC. For instance, in [8], Fraundorfer *et. al.* introduce gravity information and thereby reduces the required number of minimal samples required from 5 points to 3 points. This sample reduction improves the calculation speed exponentially. A different approach to reduce the number of required samples can be achieved by decoupling the rotation from translation (by using 2D implementation of Arun’s method) and using covariance based approach to find the relative rotation [5]. However, the latter approach neglects relative translation. These reductions can go as far as a 1 point solver which can be observed in [6]. Our work takes inspiration from the aforementioned reductions and allowing us to reduce the 5-point solver to a 3-point solver by the 2 degrees of freedom information gained through the estimated gravity vector.

3. Method

3.1. Gravity Estimation

Knowing the gravity vector of an image is an important problem for many computer vision applications and therefore much research has been done in this field. We decided to use the state-of-the-art gravity estimation model UprightNet [28] which combines local camera surface geometry and global upright surface geometry using a weighted map. This pipeline is shown in Figure 1 and Figure 2. UprightNet has an open-source implementation that also provides pre-trained weights for the ScanNet data set [4] and for the InteriorNet data set [14]. We decided to use real image data from ScanNet instead of the synthetic InteriorNet data to get realistic feedback on how our approach would work in real-world applications. UprightNet’s combination of a learning-based and a geometry-based approach results in an average angular error of 3.74° for the ScanNet data set [4]. To evaluate our pipeline, we ran the images from the ScanNet test data set through the UprightNet pipeline and re-

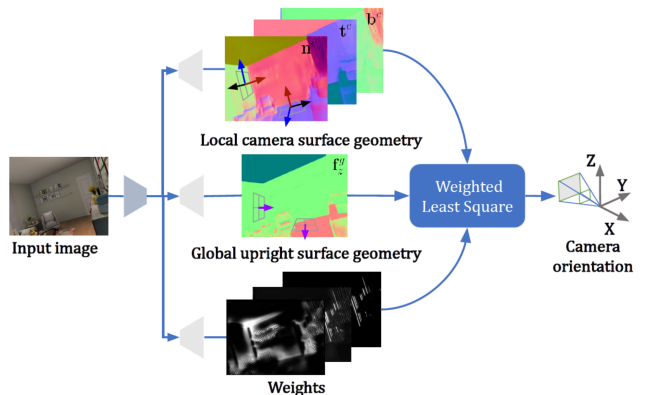


Figure 1. UprightNet model.

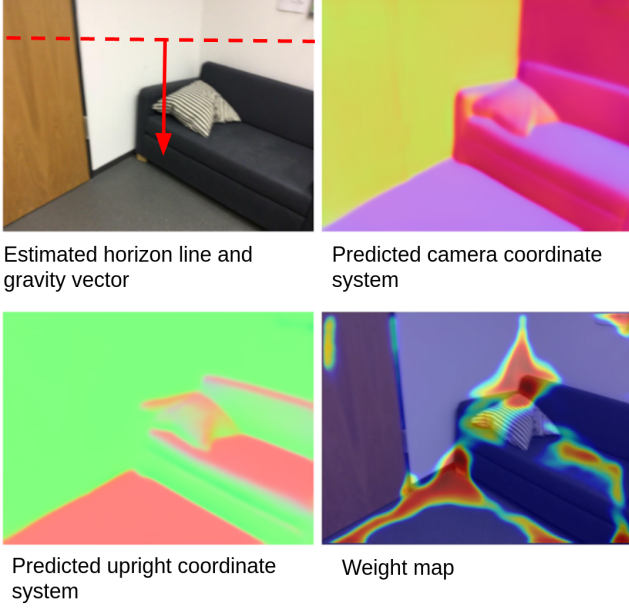


Figure 2. UprightNet gravity estimation.

ported the estimated gravity vector as well as the ground truth data. With this information, we created a dataset containing 11,434 image pairs with their corresponding intrinsic matrices, ground truth relative pose, ground truth gravity (measured using IMU) as well as the predicted gravity vector.

3.2. Pose Estimation

The pose estimation process consists of a solver for pose calculation using sampled minimal set of points, RANSAC for selecting the most plausible pose, and bundle adjustment for final relative pose refinement.

We use the *Poselib* [12] implementation of 5-point solver which is based on [19]. This method is a refinement over [20] which constructs a thirteenth degree polynomial whose roots include the possible solutions to the given 5-point problem. In contrast to [20], [19] obtains exact intrinsic degree of difficulty of the problem by using better elimination to obtain a tenth degree polynomial.

We extend *Poselib*'s 5-point estimation pipeline to create a '3-point estimator' which uses *Poselib*'s 3-point solver based on [27]. Given a known axis of rotation, [27] solves for the angle of rotation around this known axis along with the translation by formulating a *Quadratic Eigenvalue Problem*.

In our experiments, the RANSAC and bundle adjustment implementations used for both the estimators are the same. We use locally optimized RANSAC [13] implementation from *Poselib* which is based on [23]. Locally optimized RANSAC improves upon standard RANSAC by applying local optimization steps to refine the promising candidate

hypotheses generated from random minimal samples. And finally, we use bundle adjustment based on Sampson error minimization [22] for essential matrix refinement.

3.3. Metrics

To compare the two algorithms, along with runtime, we also measure the rotation and translation errors. The rotation error is the angle difference between the ground truth rotation and the estimated rotation, which is computed as $\arccos((\text{trace}(\mathbf{R}_g \mathbf{R}_e^T) - 1)/2)$, where $\mathbf{R}_g$ is the ground truth and $\mathbf{R}_e$ is the estimated rotation.

Since translation is known only up to scale, the translation error is also measured as the angle between the ground truth and the estimated translation vectors, which is obtained as $\arccos(\frac{\mathbf{t}_g}{\|\mathbf{t}_g\|} \cdot \frac{\mathbf{t}_e}{\|\mathbf{t}_e\|})$, where $\mathbf{t}_g$ and $\mathbf{t}_e$ are ground truth and estimated translations, respectively.

3.4. Pipeline

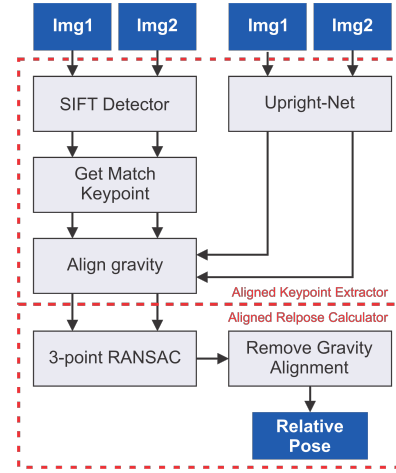


Figure 3. Relative pose estimator proposed pipeline

Figure 3 shows the whole pipeline for the 3-point estimator, which consists of 2 main modules. The first module extracts the 'aligned keypoint matches'. It finds the matched keypoints from both the input images and aligns them using the gravity vector. The second module calculates the relative pose from these aligned keypoints.

The aligned keypoint extractor takes an image pair as input with known internal calibration matrix ($\mathbf{K}$). It extracts keypoints from each image using SIFT Detector [15]. It then finds the matching keypoints using nearest neighbor with ratio check. The ratio check filters the bad matches by comparing the descriptor distance from the first best match (d_1) and the second best match (d_2). If the ratio of d_2 to d_1 is bigger than the set threshold, the matched keypoint will be rejected. In this project, we use ratio check with 0.7 as the threshold.

The gravity vector for both the images are estimated by the UprightNet (for efficiency we pre-compute and save the

estimates so that they can be reused). However, as mentioned earlier, we also test with ground truth gravity vectors and gravity estimates with synthetic noise. By using this gravity information, the matched keypoints are aligned. These aligned keypoints are then fed to the 3-point estimator to obtain the relative pose solution (in transformed frames). The final relative pose is obtained by removing the effect of alignment.

To align the keypoints, a rotation matrix $\mathbf{R}_i$ that transfers gravity vector $\hat{\mathbf{g}}_i$ to the y-axis ($\hat{\mathbf{y}}_i$) in the corresponding frame i is computed, *i.e.* $\mathbf{R}_i$ s.t. $\mathbf{R}_i \hat{\mathbf{g}}_i = \hat{\mathbf{y}}_i$; $i \in \{1, 2\}$. Here frame 1 is the normalized camera frame corresponding to the first image, relative to which we intend to compute the pose of the second one. Using this rotation matrix the keypoints ($\mathbf{x}_i$) are transformed to get the aligned keypoints ($\hat{\mathbf{x}}_i = \mathbf{R}_i \mathbf{x}_i$), which are subsequently used to obtain initial relative pose solution $[\mathbf{R}_{3pt} \mid \mathbf{t}_{3pt}]$. Then the final relative pose $[\mathbf{R}_{final} \mid \mathbf{t}_{final}]$ is obtained as follows:

$$\mathbf{R}_{final} = \mathbf{R}_2^T \mathbf{R}_{3pt} \mathbf{R}_1 \text{ and } \mathbf{t}_{final} = \mathbf{R}_2^T \mathbf{t}_{3pt}.$$

The pipeline flow for the 5-point estimator is similar except it does not have the steps involving alignment.

4. Experiments and Results

To investigate the impact of the gravity vector on the results, pose estimation was performed on 11,434 image pairs of the ScanNet data using first the ‘5-point solver’ and also the ‘3-point solver with ground truth gravity vector’. For both cases the image pairs are compared with respect to their runtime and pose errors. Image pairs with a rotational pose error larger than 20° are considered as failure cases, where the solver fails to determine the relative pose. The corresponding results are shown in Figure 4 and Table 1. From the results it is clearly visible, that the 3-point solver is not only more precise, but also faster. Compared to 5-point estimator, the mean and median rotation errors reduce by 25% while at the same time the runtime improves by 64%. Furthermore the amount of failure image pairs reduces from 22.4% to 10.7%. The time improvement can be attributed to the fact that 3-point solver needs exponentially less iterations than the 5-point solver. Furthermore it seems that knowing the gravity vector makes the estimation more stable and therefore more accurate.

Method	mean	median	failures	runtime
5pt	4.33°	2.45°	22.4%	2.64ms
3pt	3.23°	1.84°	10.7%	0.95ms
improvement	25%	25%	52%	64%

Table 1. Performance metrics 5-point estimator (5pt) vs. 3-point estimator (3pt) with IMU gravity.

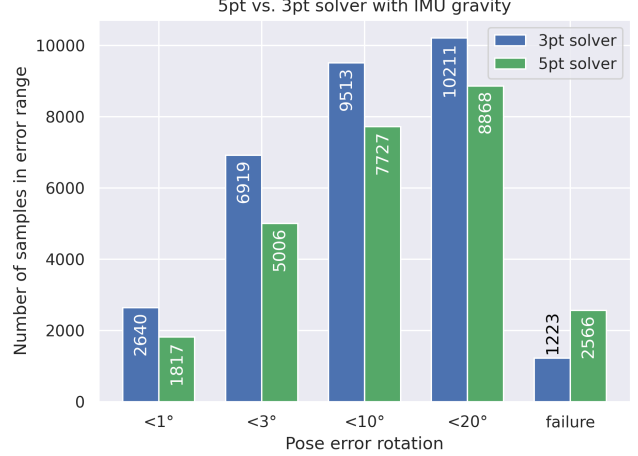


Figure 4. Performance metrics 5-point vs. 3-point solver.

We also investigated how the results change, if no IMU is available and the gravity vector has to be directly estimated from the image. Therefore we compare the results of the 3-point solver for three different cases: 1. using the ground truth (IMU) gravity vector, 2. using the UprightNet gravity estimation and 3. using a gravity vector that has the double error than the UprightNet vector. The double UprightNet error vector is obtained by calculating the angular difference α between UprightNet and ground truth gravity vector. The new vector has angular difference of 2α to the ground truth vector and lies in the same plane as the ground truth and UprightNet vector. This vector simulates a less precise gravity estimation model.

Figure 5 and Table 2 show the results of this experiment. As expected, best results are achieved when the ground truth (IMU) gravity vector is used. Using the UprightNet gravity estimation leads to less accurate results but still outperforms the 5-point estimator in terms of accuracy. If the gravity vector with double error than UprightNet is used, the results again get worse. In this case, mean and median error are also larger than the 5-point estimator. Nevertheless it is interesting to see that it still has significantly less failures than the 5-point estimator. Using UprightNet for gravity estimation leads to significantly higher runtime, as the network has a long processing time. The results show that the high accuracy of UprightNet is actually needed, to achieve significant improvement compared with the 5-point estimator. The performance metrics of UprightNet for the image pairs from our experiments are shown in Table 4.

The observed behaviour for the rotation error can also be observed for the translation error, shown in Figure 6 and Table 3. In general the translation error is significantly larger than the rotation error. This can be explained because the translation is usually very small. Even a small translation difference in the estimation can result in a large angular

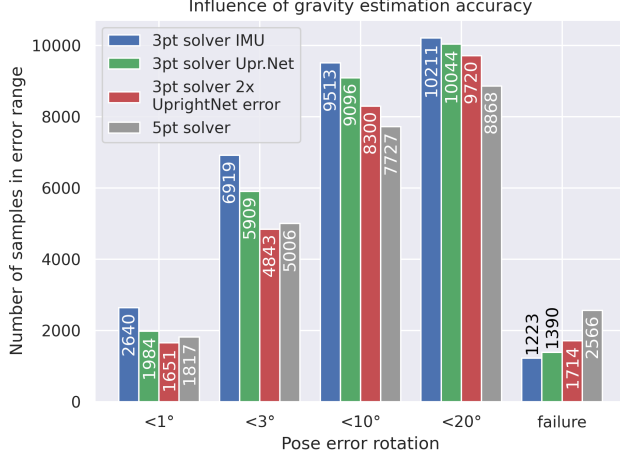


Figure 5. Influence of gravity estimation accuracy.

Gravity	mean	median	failures	runtime
Ground truth	3.23°	1.84°	10.7%	0.95 ms
UprightNet	3.90°	2.36°	12.2%	58.2 ms
2x Upr.Net error	4.78°	3.02°	14.8%	58.2 ms

Table 2. Influence of gravity estimation accuracy.

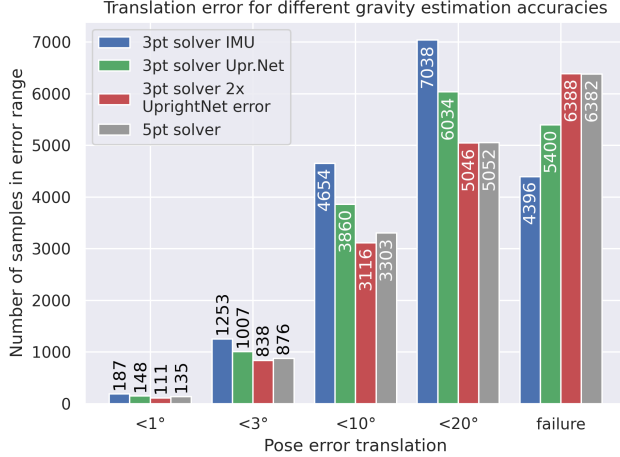


Figure 6. Translation error for different gravity estimation accuracies.

translation error. For larger displacements, the same spacial error results in a smaller angular error. Best results are again obtained with 3-point solver and ground truth (IMU) gravity.

The drawback of the 3-point solver with the gravity estimation is its long runtime. Nevertheless if the relative poses of a larger set of images has to be estimated, the runtime of the gravity estimation has less impact, as it has to be done only once for each image. For a set of n images, the gravity vector has to be estimated n times,

Method	mean	median	failures
3pt (ground truth gravity)	8.13°	7.03°	38.4%
3pt (UprightNet)	8.40°	7.46°	47.2%
3pt (2x UprighNet error)	8.65°	7.76°	55.9%
5pt	8.24°	7.21°	55.8%

Table 3. Translation errors for 5-point estimator (5pt) and 3-point estimator (3pt) with different gravity estimation accuracies.

	rotation error	roll error	pitch error
Mean	3.70°	2.10°	2.83°
Median	2.91°	1.47°	2.01°
Std.	3.56°	2.29°	3.44°

Table 4. UprightNet metrics for ScanNet test data used in our experiments.

while $\mathcal{O}(n^2)$ relative poses have to be estimated. Using larger batch sizes for UprightNet further reduces the average runtime. For example, given the average UprightNet runtime t_n of a single image in a batch of size n , the runtime for 3-point solver with UprightNet is given by $t_{3pt}(n) = n \cdot t_n + 0.95\text{ms} \cdot \frac{n(n-1)}{2}$. The runtime for 5-point solver is given by $t_{5pt}(n) = 2.64\text{ms} \cdot \frac{n(n-1)}{2}$. From our experiments, we observed that for batch size of 23, t_{23} turns out to be approximately 18 ms. For $n \geq 23$ it holds $t_{3pt}(n) < t_{5pt}(n)$. This means that for scenes with $n \geq 23$ images, 3-point solver with UprightNet is faster than the 5-point solver.

To get a understanding for the reasons when the estimation fails, failure cases were investigated. The failures can mainly be explained by a lack of good keypoints. In general, five failure categories were observed: failure due to reflection, non-overlapping images, bad lighting conditions, motion blur and repetitive structure. Examples of such cases are shown in Figure 7.

5. Limitations and Future Work

Current limitations to the project come in the form of runtime speed as well as accuracy. Since the pipeline relies on a gravity estimation component, the runtime speed is limited to the current state-of-the-art [28] which requires around 58.2 ms per estimation. However, considering the option to replace the gravity-estimation with the faster IMU-based approach, this obstacle can be overcome for implementations which integrate the required sensor readings. A further limitation comes in the form of accuracy, which relies on the robustness of keypoint detector and matching.

On a more positive note, there might be the possibility to reduce the required amount of RANSAC samples to a 1-point algorithm. This could be achieved by merging the ideas presented in this report with the rotational decoupling from translation shown in [5]. The decoupling approach

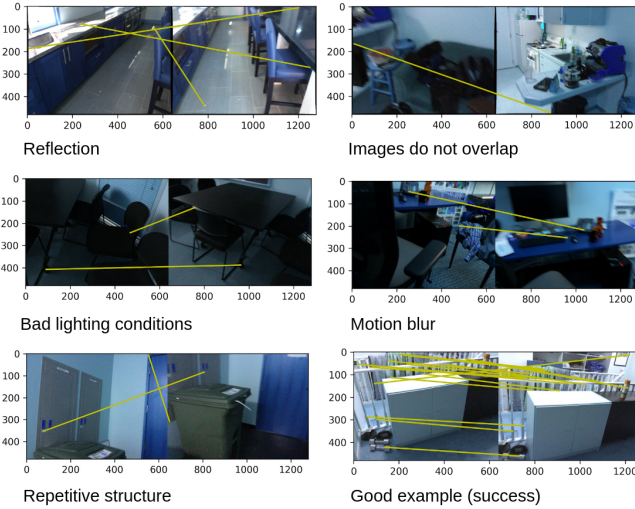


Figure 7. Five typical failure cases and one successful example.

would imply that only the relative rotation could be extracted, which neglects relative translation but still opens the door for a wide variety of use-cases within the field. A further optimization which might bring about a less restricting time limitation would be to optimize the gravity estimation by reducing the model complexity of UprightNet [28].

6. Conclusion

The 3-point algorithm promises better performance than the current 5-point algorithm, both in accuracy and speed. The proposed method is limited by the computational complexity of the gravity estimation which can be overcome by using IMU data. However, for applications involving large number of image pairs, 3-point algorithm with estimated gravity is still faster than the 5-point algorithm. Hence, the proposed pipeline could find a use case for computing relative poses with higher precision and speed. As a last note, it must be stated that this pipeline was solely tested on an indoor dataset [4] and may not prove to be as effective on other datasets and scenes.

7. Implementation

- ‘3-point estimator class’ and corresponding *pybind*, which combines *Poselib*’s existing 3-point solver and RANSAC implementation were implemented by us. 5-point estimator was directly taken from *Poselib*¹.
- The **comparison pipeline** was fully implemented by us which includes data pre-processing, synthetic data generation, keypoint alignment & de-alignment, metrics and benchmarking *etc.*

¹<https://github.com/vlarrsson/PoseLib>

- UprightNet implementation² was analyzed and adapted to report the estimated gravity vector for each image and add the results to the corresponding data folder.

8. Work Distribution

- UprightNet Gravity Estimation (Nicola, Timo) (*Code repository*: <https://github.com/burgern/UprightNet>)
 - Setup on Euler platform (Nicola)
 - Investigate ScanNet data, train own model and compare to pre-trained weights (Nicola, Timo)
 - Adapt pipeline to add estimated gravity vectors to data folders (Timo)
- 3-point estimator and *pybind* (Shubham, Yoga) (*Code repository*: <https://gitlab.ethz.ch/kumarsh/poselib-3dv>)
- Comparison pipeline (*Code repository*: <https://gitlab.ethz.ch/kumarsh/3d-vision>)
 - Dataset loading and standardization (Nicola, Timo)
 - Synthetic data for verification (Shubham, Yoga)
 - Keypoint matching (Yoga)
 - Pose processing (Shubham, Yoga)
 - Alignment (Shubham, Yoga)
 - Angle Interpolation (Timo, Yoga)
 - Metrics (Shubham)
 - Benchmarking (Shubham, Yoga)
 - Reporting (Nicola, Shubham, Timo, Yoga)
 - Visualization (Nicola, Shubham)
- Presentation, Poster, Report (Nicola, Shubham, Timo, Yoga)

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²<https://github.com/zhengqili/UprightNet>

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